

# Muhammad Arham

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## EDUCATION

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**National University of Computer And Emerging Sciences, Chiniot Faisalabad Campus**

Bachelor of Science, Computer Science - 2026

(3.59 / 4)

**Nixor College Karachi**

Alevels - Computer Science - 2022

4A\*

## PROJECTS

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### SoccerVision

- Developed an end-to-end AI system for automated football match highlight generation
- Implemented computer vision models to detect key events (goals, penalties, freekicks etc.)
- Generated multiple highlight types (full match, event-based, match summaries) from raw footage
- Integrated APIs for payments, authentication and social media sharing.

### QLoRA Fine-Tuning of Vision Language Models

- Fine-tuned the vision-language model Qwen Qwen2-VL-2B-Instruct using QLoRA for document image to Markdown conversion.
- Applied parameter-efficient fine-tuning techniques including 4-bit quantization, LoRA adapters, gradient accumulation, and GPU-based training on Kaggle.

### Masked Auto Encoder (MAE)

- Built a self-supervised MAE using ViT (Base + Small) for image reconstruction
- Applied 75% patch masking and asymmetric encoder-decoder design
- Optimized training with mixed precision on dual T4 GPUs (Kaggle)

### Custom Diffusion Model for Image Generation

- Implemented a Denoising Diffusion Probabilistic Model (DDPM) in PyTorch using a custom U-Net architecture for high-resolution image generation and reconstruction from noisy inputs.
- Developed the complete forward and reverse diffusion pipeline, including noise scheduling, timestep embeddings, mixed-precision training, and image quality evaluation using PSNR and SSIM metrics

### Image Captioning System

- Built a multimodal image captioning model using ResNet50 + LSTM (Seq2Seq)
- Extracted and cached 2048-d image features to optimize training

### Traffic Sign Detection & Classification

- Built a computer vision pipeline using YOLO and CNN models for detection and classification tasks
- Fine Tuned models on traffic signs dataset

## Activites, Honour and Awards

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6x Deans List, FAST NUCES Chiniot-Faisalabad

AI & Computer Network TA, FAST NUCES Chiniot-Faisalabad

99.3 Percentile in NSCT TEST, NUCES, HEC

3<sup>rd</sup> Position FPSC coding competition, FAST NUCES Chiniot-Faisalabad

## TECHNICAL EXPERTISE

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**Languages:** Python, C++, SQL

**Frameworks & Libraries:** PyTorch, NumPy, FastAPI, Scikit-learn, LangChain

**Tools & Platforms:** Git, Docker, AWS, Ollama, Kaggle, Firebase

## Interests & Hobbies

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GeoPolitics, History, Sports